

i. Production of beats:

a. When there is the superposition of two sound waves having the same

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production of beats.

b. The variation in the loudness of sound that goes up and down is the phenomenon of the formation of beats.

c. It can be considered as the superposition of waves and the formation of standing waves in time at one point in space where waves of slightly different frequencies are passing. The two waves are in and out of phase giving constructive and destructive interference.

ii. Expression for beats frequency:

a. Consider two sound waves, having the same amplitude and slightly different frequencies n_1 and n_2 .

b. Let x be a point of the medium where they arrive in phase.

c. The displacement due to each wave at any instant of time at that point is given as

$$y_1 = a \sin \left[2\pi \left(n_1 t - \frac{x}{\lambda_1} \right) \right]$$

$$y_2 = a \sin \left[2\pi \left(n_2 t - \frac{x}{\lambda_2} \right) \right]$$

d. Let us assume for simplicity that the listener is at $x = 0$.

$$\therefore y_1 = a \sin(2\pi n_1 t) \text{ and } y_2 = a \sin(2\pi n_2 t)$$

e. According to the principle of superposition of waves,

$$y = y_1 + y_2$$

$$\therefore y = a \sin(2\pi n_1 t) + a \sin(2\pi n_2 t)$$

$$\therefore y = 2a \sin \left[2\pi \left(\frac{n_1 + n_2}{2} \right) t \right] \cos \left[2\pi \left(\frac{n_1 - n_2}{2} \right) t \right]$$

$$[\text{By using formula, } \sin C + \sin D = 2 \sin \left(\frac{C + D}{2} \right) \cos \left(\frac{C - D}{2} \right)]$$

Rearranging the above equation, we get

$$y = 2a \cos \left[\frac{2\pi(n_1 - n_2)}{2} t \right] \sin \left[\frac{2\pi(n_1 + n_2)}{2} t \right]$$

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$$\therefore y = A \sin(2\pi n t)$$

This is the equation of a progressive wave having frequency 'n' and amplitude 'A'. The frequency 'n' is the mean of the frequencies n_1 and n_2 of arriving waves while the amplitude A varies periodically with time.

g. The intensity of sound is proportional to the square of the amplitude. Hence the resultant intensity will be maximum when the amplitude is maximum.

h. For maximum amplitude (waxing),

$$A = \pm 2a$$

$$\therefore 2a \cos \left[\frac{2\pi(n_1 - n_2)}{2} t \right] = \pm 2a$$

$$\therefore \cos \left[\frac{2\pi(n_1 - n_2)}{2} t \right] = \pm 1$$

$$\text{i.e., } \left[2\pi \left(\frac{n_1 - n_2}{2} \right) t \right] = 0, \pi, 2\pi, 3\pi, \dots$$

$$\therefore t = 0, \frac{1}{n_1 - n_2}, \frac{2}{n_1 - n_2}, \frac{3}{n_1 - n_2}, \dots$$

Thus, the time interval between two successive maxima of sound is always

$$\frac{1}{n_1 - n_2}$$

$$\text{Hence, the period of beats is } T = \frac{1}{n_1 - n_2}$$

The number of waxing heard per second is the reciprocal of period of waxing.

$$\therefore \text{Beat frequency in waxing, } N = n_1 - n_2$$

iii. The intensity of the sound will be minimum when the amplitude is zero (waning):

For minimum amplitude, $A = 0$,

$$\therefore 2a \cos \left[2\pi \left(\frac{n_1 - n_2}{2} \right) t \right] = 0$$

$$\therefore \cos \left[2\pi \left(\frac{n_1 - n_2}{2} \right) t \right] = 0$$

$$\therefore \left[2\pi \left(\frac{n_1 - n_2}{2} \right) t \right] = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

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The number of waning heard per second is the reciprocal of period of waning.

Period of beat, $T = \frac{1}{n_1 - n_2}$

Beat frequency in waning, $N = n_1 - n_2$



Concept: [Beats](#)

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